

WEEK 5 - ReactJS Hands ON - 7995206

Exercise 1

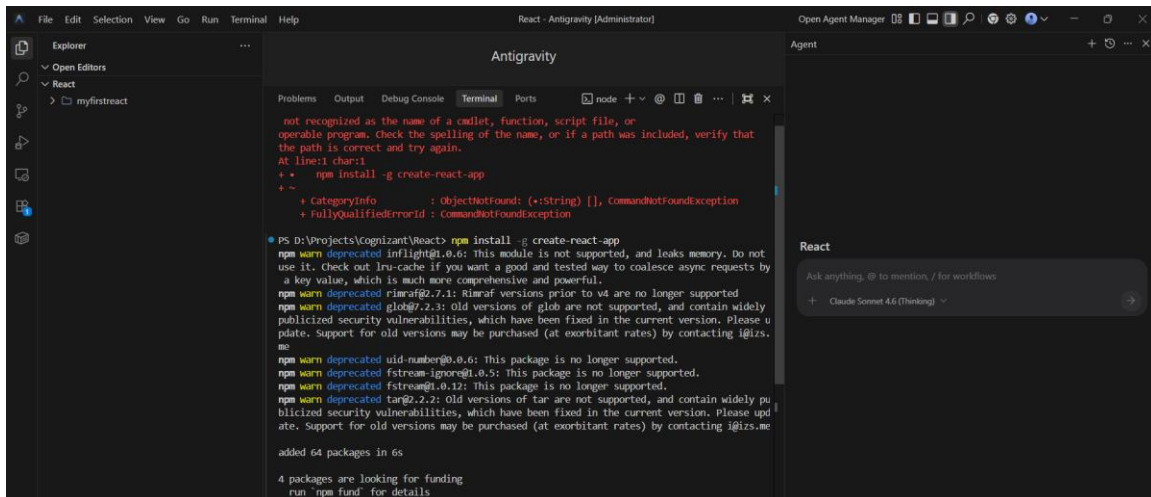
Create myfirstreact app and display 'Welcome to the first session of React'.

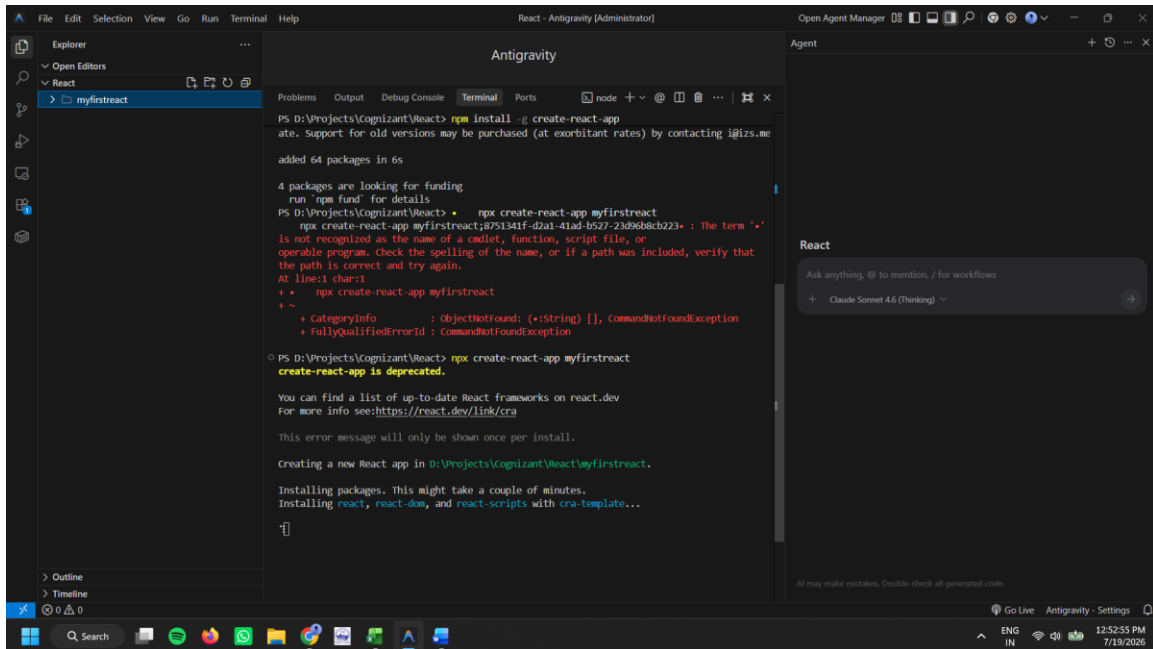
Commands

- npm install -g create-react-app
- npx create-react-app myfirstreact
- cd myfirstreact
- npm start

Solution:

```
function App() {  
  return <h1>Welcome to the first session of React</h1>;  
}  
export default App;
```





Exercise 2

Class Components

Aim:

StudentApp with Home, About and Contact components.

Commands

- `npx create-react-app StudentApp`
- `cd StudentApp`
- `npm start`

Sample Code

```
// Home.js
import React, {Component} from 'react';
export default class Home extends Component{
  render(){
    return <h3>Welcome to the Home Page of Student Management
Portal</h3>;
  }
}
```

Exercise 3

Functional Components

Aim:

Score Calculator using props and CSS.

Commands

- `npx create-react-app scorecalculatorapp`
- `npm start`

Sample Code

```
export function CalculateScore({Name, School, total, goal}) {  
  const score=((total/goal).toFixed(2));  
  return (  
    <div>  
      <h1>Student Details</h1>  
      <p>Name: {Name}</p>  
      <p>School: {School}</p>  
      <p>Total: {total}</p>  
      <p>Score: {score}%</p>  
    </div>  
  );  
}
```

Exercise 4

Lifecycle Methods

Aim:

Create a React application that fetches blog posts using `componentDidMount()` and handles errors using `componentDidCatch()`.

Commands

- `npx create-react-app blogapp`
- `cd blogapp`
- `npm start`

Post.js

```
class Post {
```

```
constructor(id, title, body) {  
  this.id = id;  
  this.title = title;  
  this.body = body;  
}  
}
```

```
export default Post;
```

Posts.js

```
import React, { Component } from "react";  
import Post from "./Post";
```

```
class Posts extends Component {  
  constructor(props) {  
    super(props);  
    this.state = { posts: [] };  
  }
```

```
  loadPosts() {  
    fetch("https://jsonplaceholder.typicode.com/posts")  
      .then((res) => res.json())  
      .then((data) => {  
        const posts = data.map(  
          (item) => new Post(item.id, item.title, item.body)  
        );  
        this.setState({ posts });  
      });  
  }
```

```
});  
}
```

```
componentDidMount() {  
  this.loadPosts();  
}
```

```
componentDidCatch(error) {  
  alert("Something went wrong!");  
  console.log(error);  
}
```

```
render() {  
  return (  
    <div>  
      <h1>Blog Posts</h1>  
      {this.state.posts.map((post) => (  
        <div key={post.id}>  
          <h3>{post.title}</h3>  
          <p>{post.body}</p>  
          <hr />  
        </div>  
      ))}  
    </div>  
  );  
}
```

```
}
```

```
export default Posts;
```

App.js

```
import Posts from "./Posts";
```

```
function App() {
```

```
  return <Posts />;
```

```
}
```

```
export default App;
```

Exercise 5

CSS Modules & Styling

Aim

Style the React application using **CSS Modules**. Create a CohortDetails.module.css file and apply styles to display ongoing and completed cohorts.

Commands

```
npm install
```

```
npm start
```

Folder Structure

```
src
```

```
|
```

```
├── App.js
```

```
├── CohortDetails.js
```

└─ CohortDetails.module.css

└─ index.js

CohortDetails.module.css

```
.box {  
  width: 300px;  
  display: inline-block;  
  margin: 10px;  
  padding: 10px 20px;  
  border: 1px solid black;  
  border-radius: 10px;  
}
```

```
dt {  
  font-weight: 500;  
}
```

CohortDetails.js

```
import styles from "./CohortDetails.module.css";
```

```
function CohortDetails(props) {  
  
  return (  
  
    <div className={styles.box}>
```

```
<h3
  style={{
    color:
      props.status === "ongoing"
        ? "green"
        : "blue"
  }}
>
  {props.name}
</h3>
```

```
<dl>
```

```
  <dt>Started On</dt>
```

```
  <dd>{props.startDate}</dd>
```

```
  <dt>Current Status</dt>
```

```
  <dd>{props.status}</dd>
```

```
  <dt>Coach</dt>
```

```
  <dd>{props.coach}</dd>
```

```
  <dt>Trainer</dt>
```

```
  <dd>{props.trainer}</dd>
```

```
</dl>
```



```
</div>
```

```
);
```

```
}
```

```
export default CohortDetails;
```

App.js

```
import CohortDetails from "../CohortDetails";
```

```
function App() {
```

```
  return (
```

```
    <div>
```

```
      <h1>My Academy Cohorts</h1>
```

```
      <CohortDetails
```

```
        name="React JS"
```

```
        startDate="10-Jan-2025"
```

```
        status="ongoing"
```

```
        coach="John"
```

```
        trainer="David"
```

```
/>
```

```
<CohortDetails
```

```
  name="Java Full Stack"
```

```
  startDate="15-Dec-2024"
```

```
  status="completed"
```

```
  coach="Smith"
```

```
  trainer="Alex"
```

```
/>
```

```
<CohortDetails
```

```
  name="Python"
```

```
  startDate="20-Feb-2025"
```

```
  status="ongoing"
```

```
  coach="Emma"
```

```
  trainer="Sophia"
```

```
/>
```

```
</div>
```

```
);
```

```
}
```

```
export default App;
```

index.js

```
import React from "react";

import ReactDOM from "react-dom/client";

import App from "./App";

const root = ReactDOM.createRoot(
  document.getElementById("root")
);

root.render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```